BACHELOR OF COMPUTER SCIENCE CAPSTONE DEFENSE

Design and Implementation of an AI-Enhanced Restaurant Management System

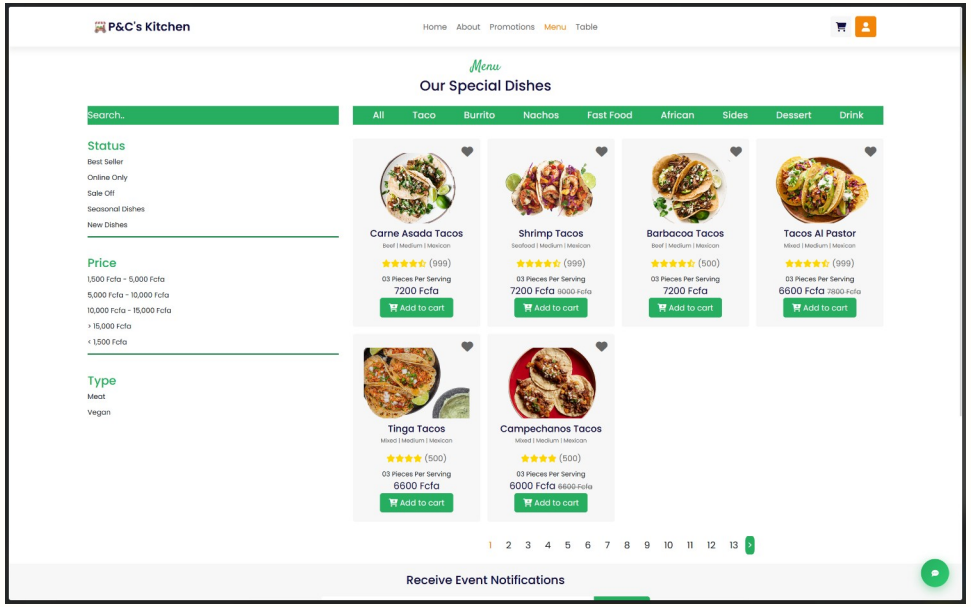
A full restaurant platform with online ordering, admin operations, recommendations, NLP assistance, and demand forecasting.

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CONTEXT

The problem is operational, not only technical

- Customers need faster ways to browse, choose, order, and follow up.
- Restaurant staff need clearer visibility over orders, payment, and delivery status.
- Managers need demand signals before purchasing ingredients or planning work.
- The project connects daily restaurant operations with AI-assisted support.



Core point

The system solves an operational workflow first. The AI features strengthen that workflow where they are useful.

OBJECTIVE

The project objective is a working restaurant platform

Customer side

Browse the menu, view food details, add items to cart, checkout, book tables, and track orders.

Restaurant side

Receive orders, update order status, manage payment and delivery progress, and monitor demand forecasts.

AI assistance

Personalize recommendations, answer menu questions through NLP, and forecast demand from completed orders.

The system is not an isolated AI demo. It is a restaurant management workflow with AI embedded where it can reduce friction and support decisions.

SYSTEM SCOPE

The website covers the main restaurant journey

Customer modules

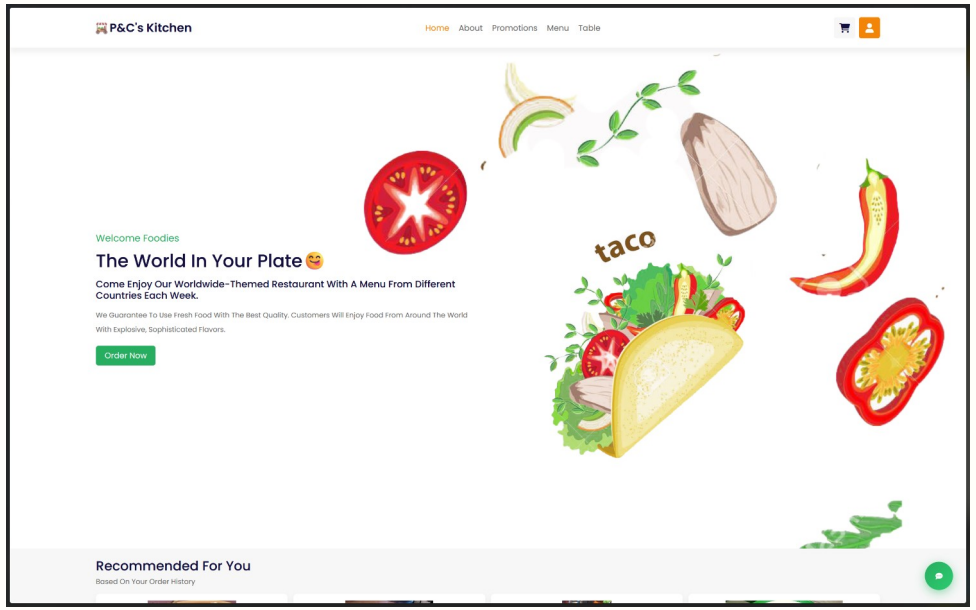
Home, 74-dish multi-cuisine menu (Mexican, Ivorian, Fast Food), promotions, cart with live item badge, user profile, checkout, table booking, and order tracking.

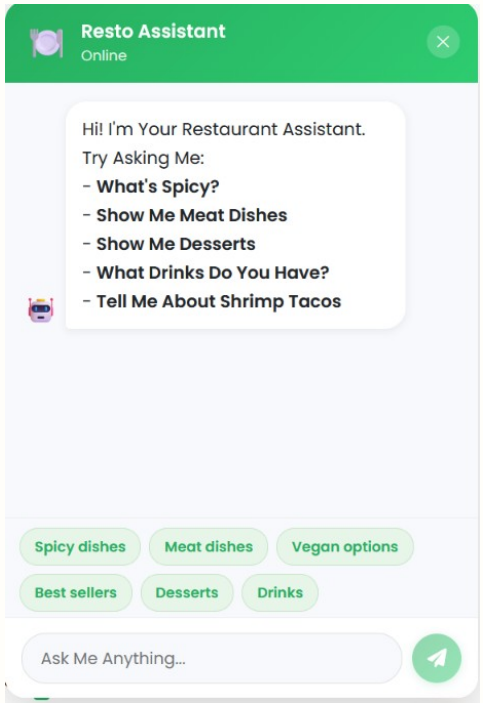
Admin modules

Admin login, order list, status updates, payment actions, and delivery control.

AI modules

Recommendation system, chatbot assistant, and demand forecasting service.





The architecture separates interface, logic, and data

01

Vue.js frontend

02

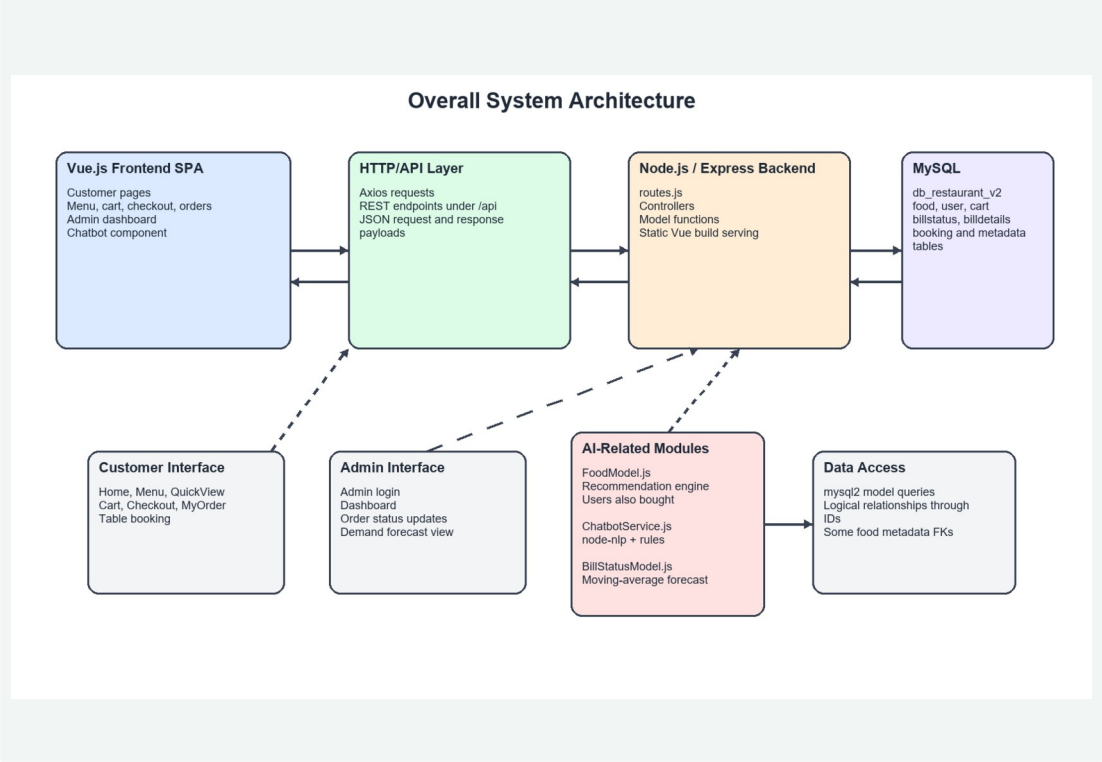
Node/ Express API

03

MySQL database

AI service layer

Recommendation logic, chatbot intent handling, and moving-average forecasting are connected through REST endpoints and database records.



The customer workflow supports ordering from discovery to checkout

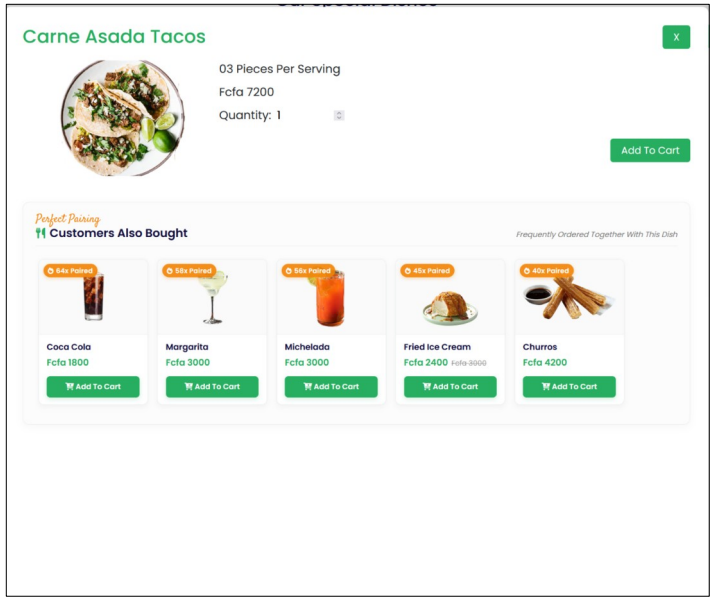
01
Browse menu

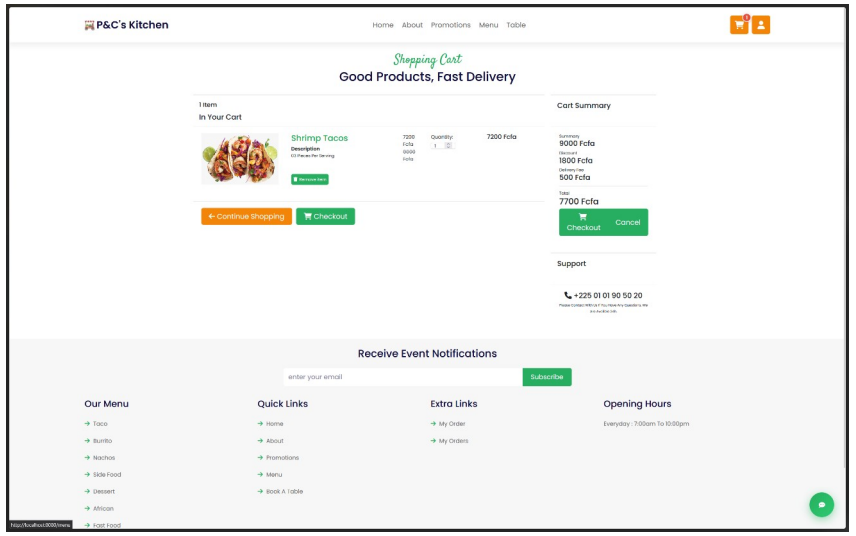
02
View food

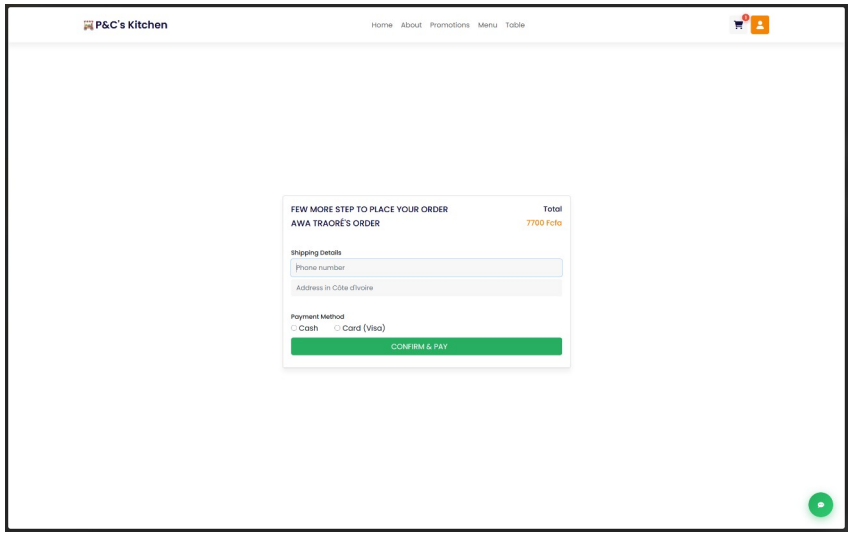
03
Add to cart

04
Checkout

05
Track order

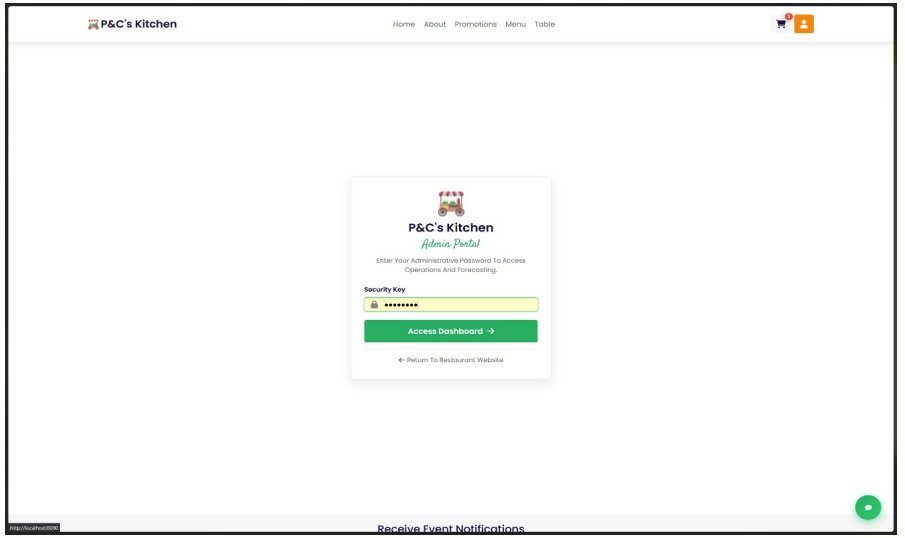


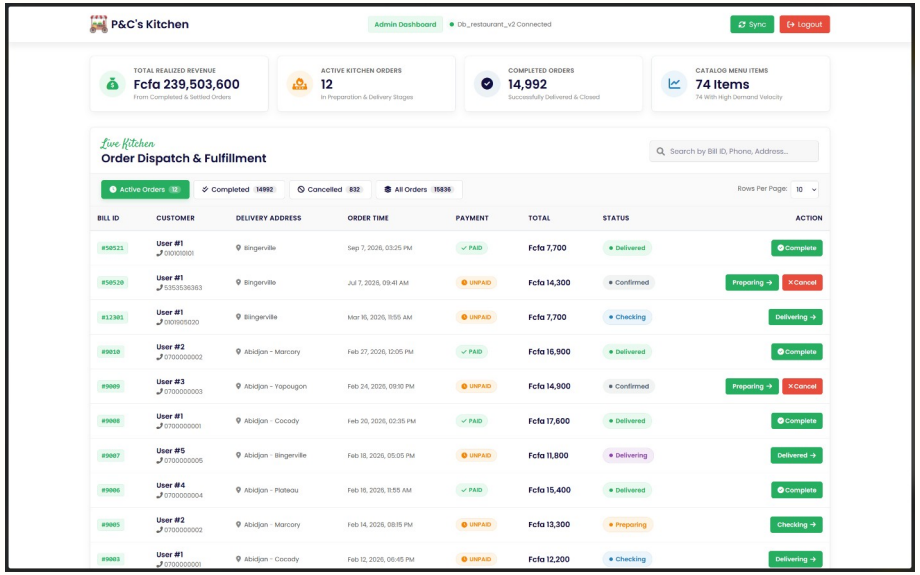




The admin workflow turns orders into managed operations

- The dashboard lists active orders and customer details.
- Status controls move orders through preparation and delivery.
- Payment and cancellation actions support operational control.
- Forecasting appears inside the same admin workspace.





AI improves specific points in the restaurant workflow

Choosing food

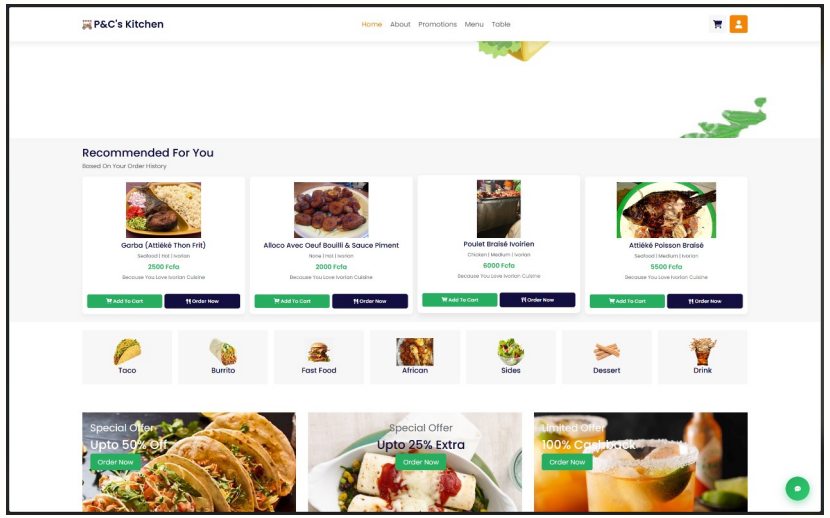
Hybrid recommender combines cuisine affinity (weight 50), category (25), food type (10), and sales popularity with explainable suggestions.

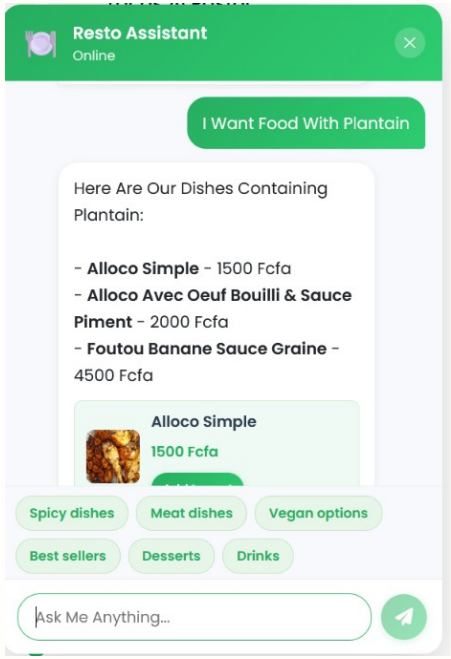
Asking naturally

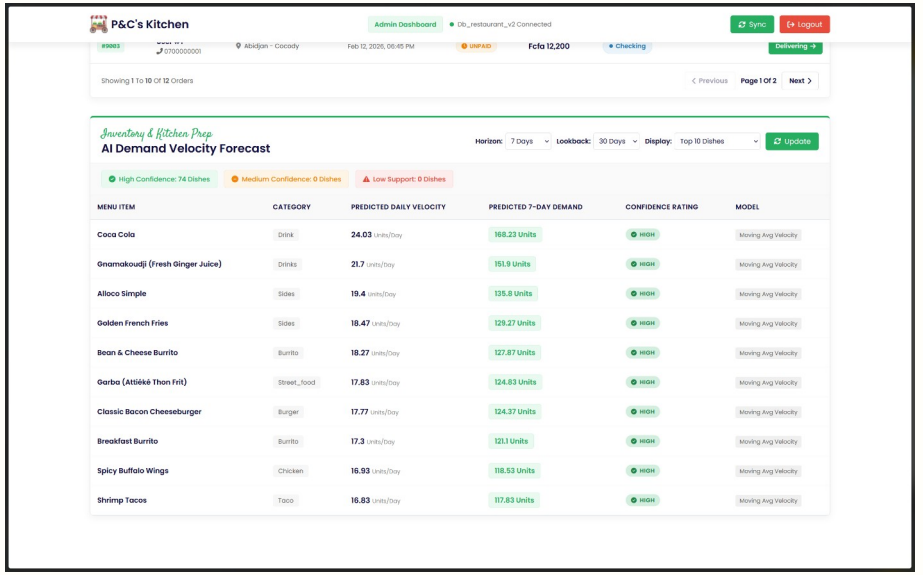
Chatbot maps customer queries across core intents (recommendations, spicy, vegan, drinks, ingredients) and returns database-aware answers.

Demand forecasting

Our run-rate model analyzes 30 days of daily sales to anticipate 7-day ingredient needs and guide restocking.



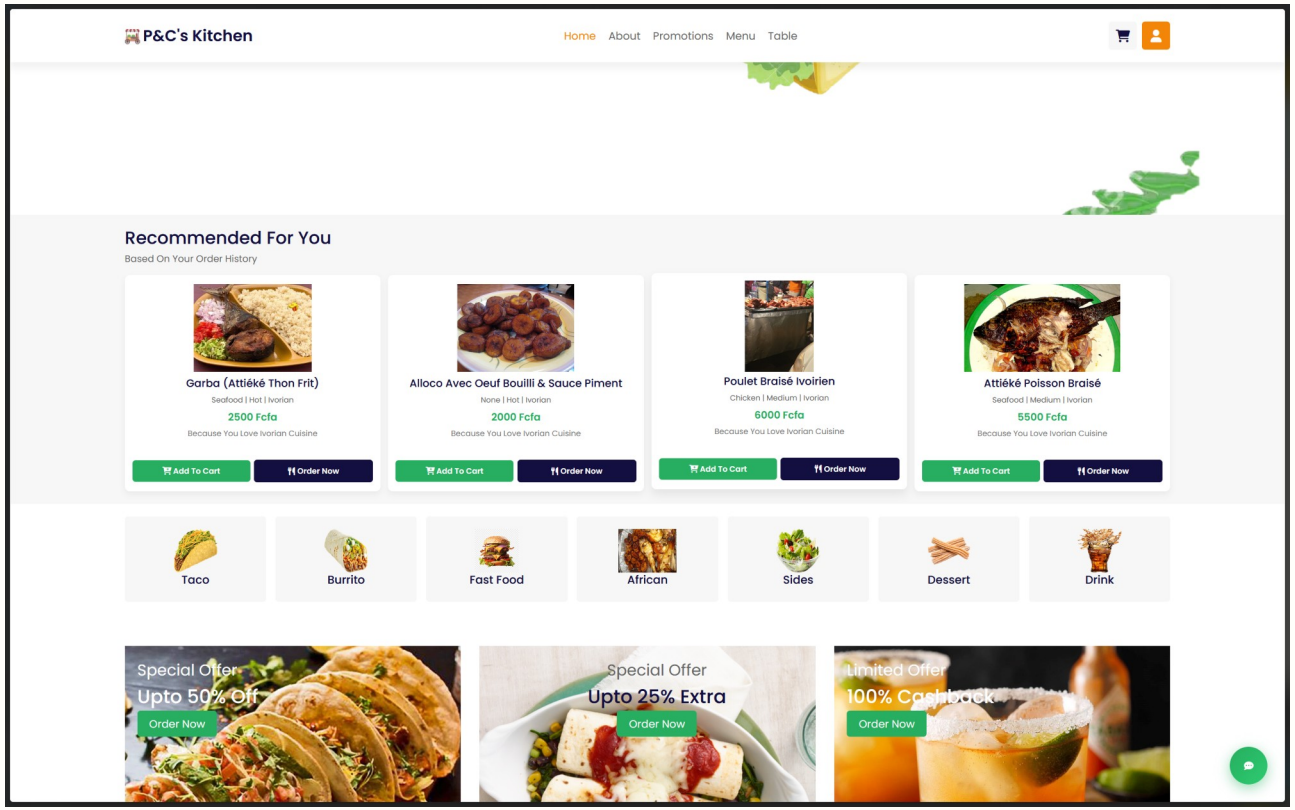




Recommendations personalize the menu without hiding popular items

To help our customers navigate a large 74-item menu, we built an AI recommendation system. Instead of just showing random items, the system analyzes a customer's past orders to understand their unique taste profile

It combines these personal affinities with global trends, ensuring customers receive tailored suggestions while still discovering our restaurant's best-sellers."



Technical Deep Dive: Hybrid Recommendation Engine

Algorithmic Approach:

We built a Hybrid Engine combining content-based filtering with collaborative signals.

The Math:

The core is our scoring formula: Score = (50 × Cuisine Weight) + (25 × Category Weight) + (10 × Type Weight) + Normalized Popularity.

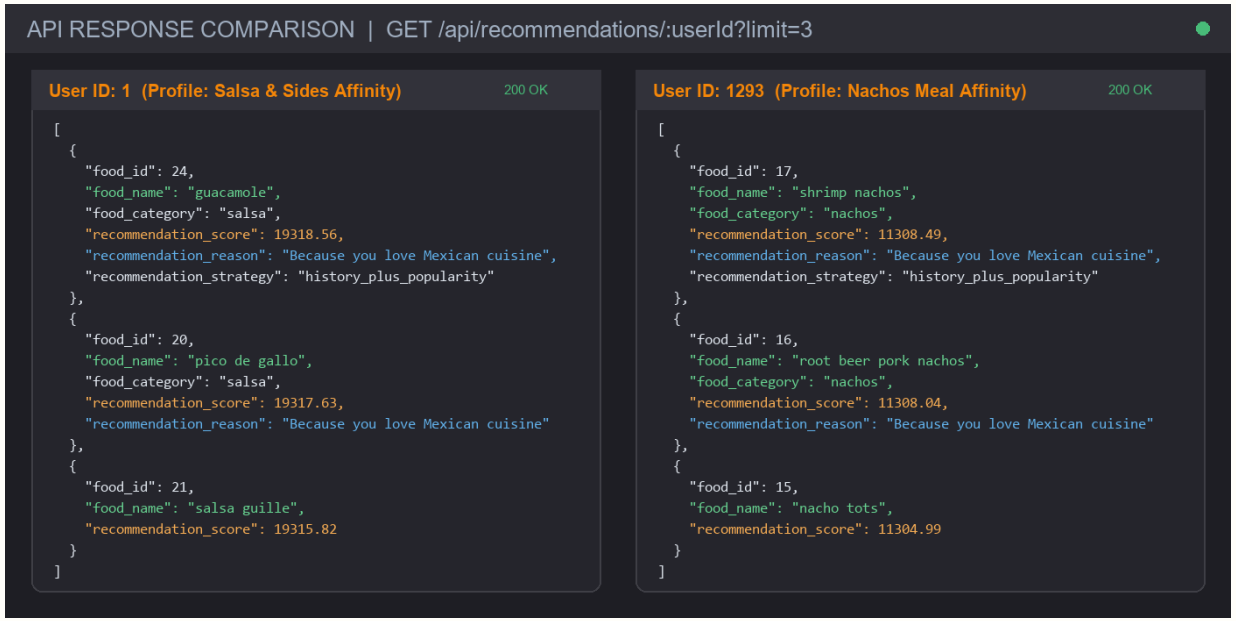
Cross-Selling:

For the Market Basket Analysis, we execute a SQL Self-Join on the billdetails table to extract exact co-purchase frequencies.

Metrics & Evaluation:

We evaluated the model offline on 298 users, achieving a Precision@1 of 0.634 and a Recall@5 of 0.536.

Limitation: The computational complexity is O(N) which scales well, but the strict limitation is the cold-start problem, requiring a fallback to global popularity for new users.

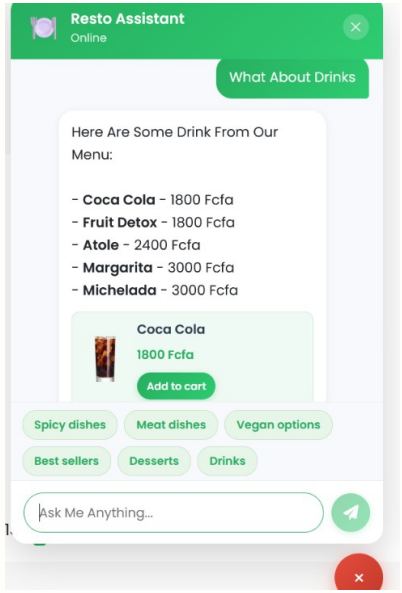


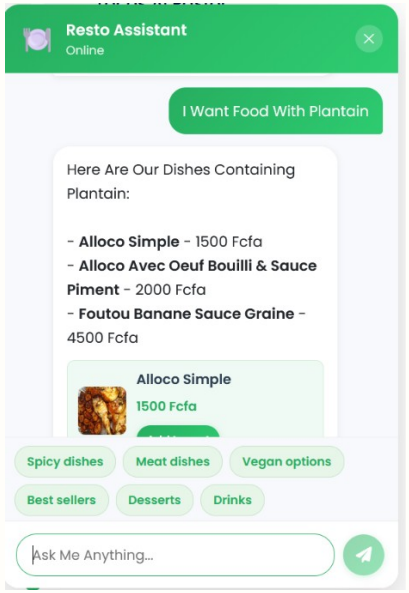
0.634
Precision@1

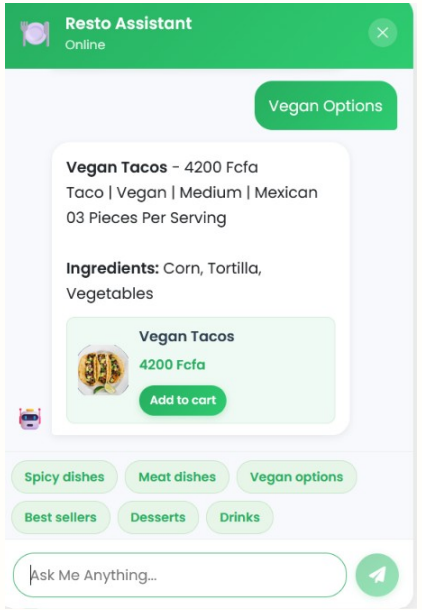
0.536
Recall@5

The chatbot turns customer language into restaurant actions

Clicking through complex menus can cause friction, so we introduced an NLP Chatbot to act as a virtual waiter. Customers can ask natural questions like 'What vegan food do you have?' or 'What are the ingredients in the Garba?'. The bot understands their intent, queries the database, and instantly returns interactive food cards, making the ordering process conversational and immediate.







Technical Deep Dive: NLP Chatbot Pipeline

NLP Pipeline:

Text -> Stemming -> Intent Match -> Entity Extraction -> SQL Query.

Mathematical Model:

The core classifier is a Logistic Regression model, trained entirely in-house on a supervised dataset of 225 utterance pairs across 10 distinct intents.

Entity Resolution:

We use string matching to link extracted tokens directly to the explicit food_id in the MySQL database.

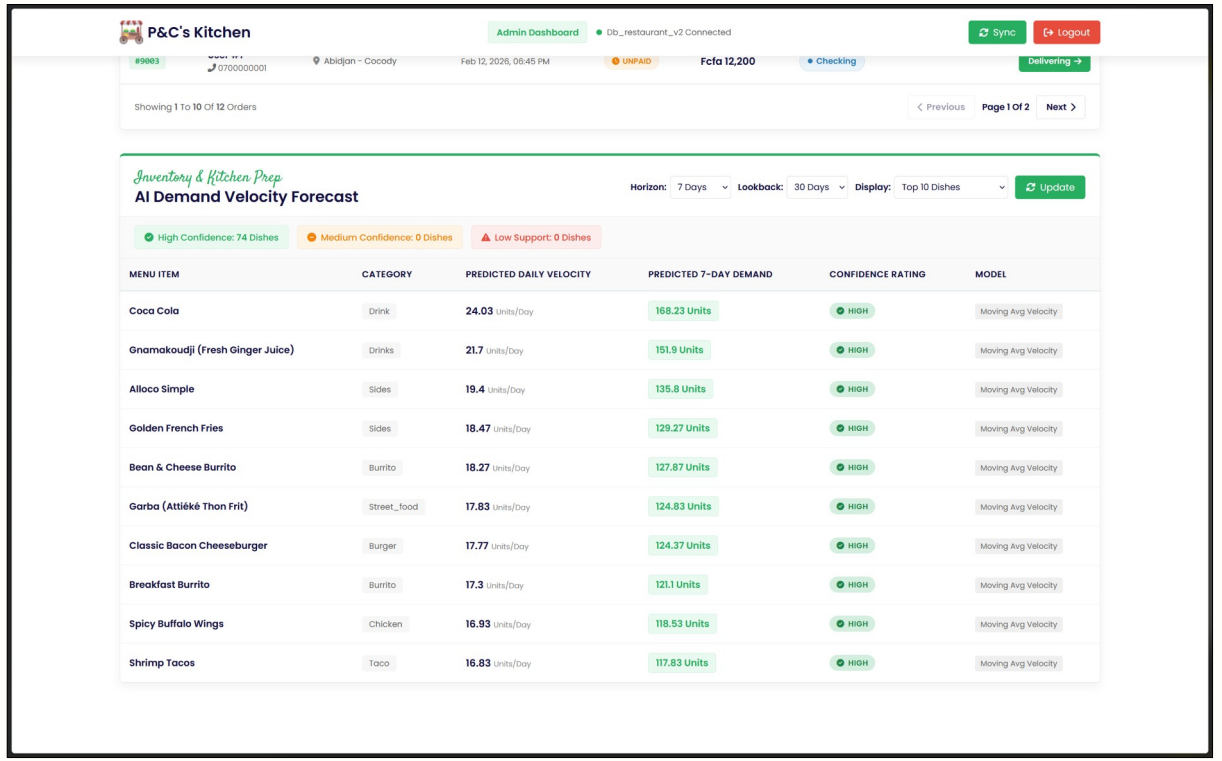
Limitation:

Being 100% in-house means the routing is deterministic. It cannot generate novel conversational text like an LLM, but this architectural choice guarantees zero data leaks and prevents AI hallucinations.

Forecasting gives the admin a practical planning signal

Our AI Demand Forecasting engine predicts future ingredient needs by learning from historical sales velocities. Integrated directly into the admin dashboard, it bridges daily transactions with smart kitchen inventory planning:

- **The challenge:** managing perishable ingredients across a 74-dish menu is difficult without clear visibility into upcoming demand.
- **The solution:** a moving-average daily velocity model that continuously tracks completed orders in our MySQL database.
- **Proactive restocking:** it calculates the units needed for the next 7 days, so managers order the right quantities and cut waste.
- **Actionable signals:** every dish gets a confidence tag, so the kitchen knows instantly which forecasts are the most reliable.



Technical Deep Dive: Demand Forecasting Model

The Model:
We designed a moving-average time-series run-rate model that runs deterministically in O(N) time, straight from our relational database.

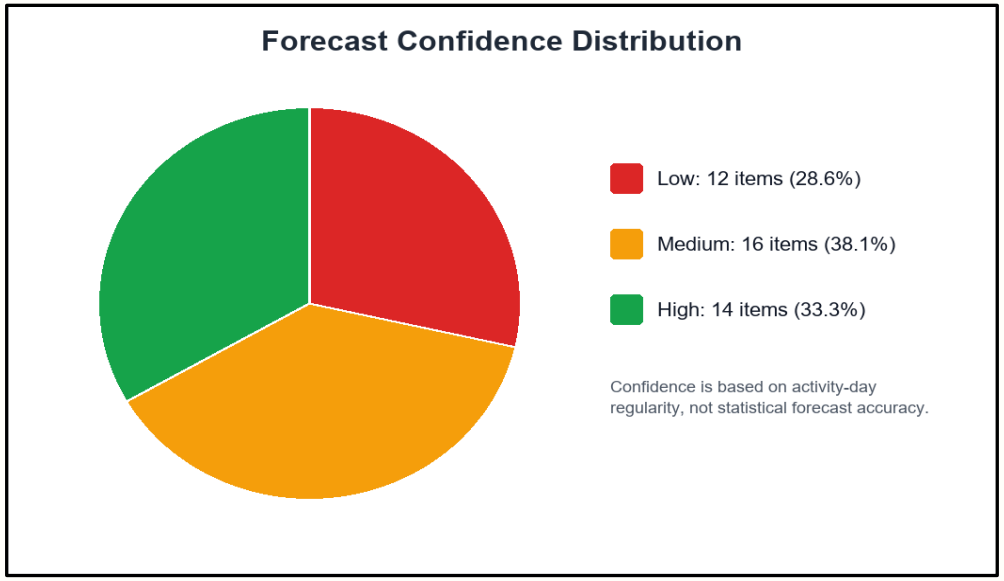
How It Calculates:
It divides the units sold over a 30-day rolling window by 30 to get a daily velocity, then multiplies that rate by 7 to project weekly demand.

Reliability Metric:
An Activity Ratio (active sales days divided by 30) ranks each dish as High (above 50%), Medium (above 20%) or Low confidence.

Model Characterization:
Forecasting is a continuous regression task, not a classification one, so the model is judged on sales regularity and volume stability rather than precision or recall.

Limitation:
As a purely historical average, it cannot anticipate sudden shocks such as severe weather or unexpected holidays.

14	16	12
High confidence	Medium confidence	Low confidence



Mathematical Formulation & Proof

- **Daily Sales Velocity:** $V_{daily} = (1 / W) \times \Sigma Qty_t$ (Lookback $W = 30$ days)
- **7-Day Demand Forecast:** Forecast $Y_{7d} = V_{daily} \times H$ (Horizon $H = 7$ days)
- **Activity Regularity Ratio:** AR = Active Sales Days / 30 $[0 \le AR \le 1]$
- **Confidence Thresholds:** High: $AR \ge 50\%$ | Medium: $AR \ge 20\%$ | Low: $AR < 20\%$

Evaluation shows progress while staying honest about scope

Recommendation

298 evaluated users. Precision@1: 0.6342. Recall@5: 0.5360.

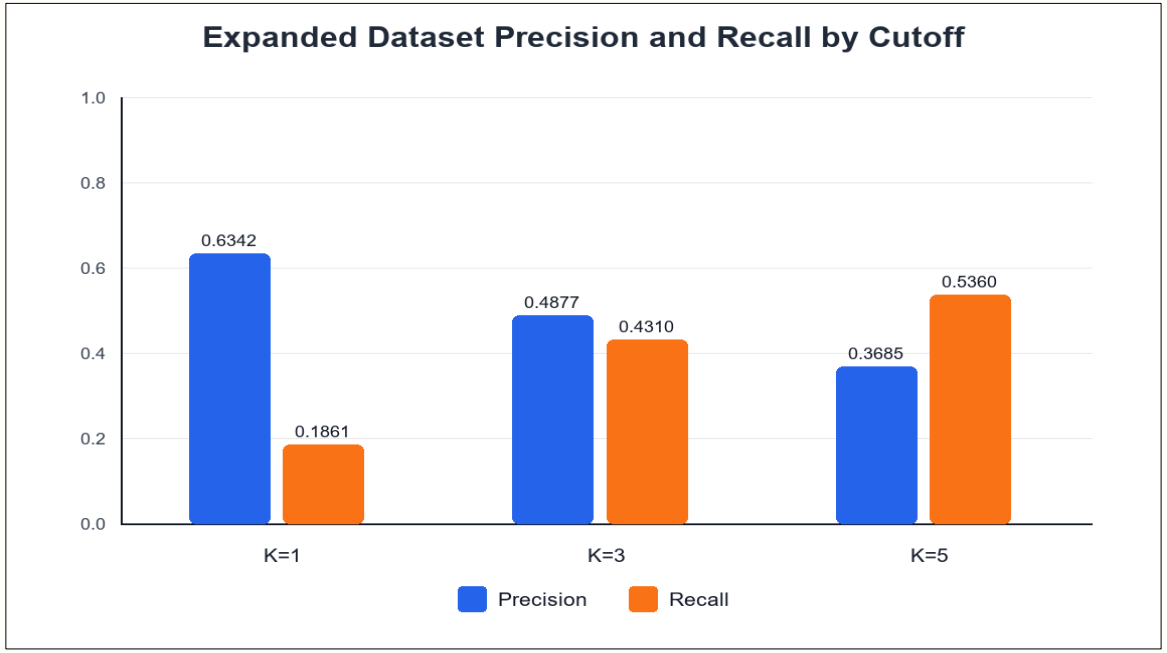
Chatbot

64 validation utterances across 8 core intents (93.75% accuracy), extended to 10 intents in live chat.

Forecast

Tested on 42 baseline items (14 High, 16 Medium, 12 Low) and scaled to all 74 live dishes over 5,000+ orders.

Evaluated in a development environment on 298 users, 64 chatbot test utterances and 5,000+ order records.



CONCLUSION

The result is a full restaurant system with AI support

- Delivers a complete, ergonomic restaurant application (74 items, user profiles, real-time cart badge, admin dashboard).
- Seamlessly embeds 3 AI capabilities: personalized hybrid recommendations, NLP chatbot, and inventory demand forecasting.
- Full-stack implementation: Vue.js frontend, Node.js/Express REST API, and MySQL relational database.
- Thoroughly evaluated offline benchmarks paired with production-ready live interactive features.

Core contribution

A working restaurant management website where AI improves everyday customer and admin decisions.

THANK YOU FOR YOUR ATTENTION